

Resilient Control Design Based on a Sampled-Data Model for a Class of Networked Control Systems Under Denial-of-Service Attacks

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Abstract—This article is concerned with designing resilient state feedback controllers for a class of networked control systems under denial-of-service (DoS) attacks. The sensor samples system states periodically. The DoS attacks usually prevent those sampled signals from being transmitted through a communication network. A logic processor embedded in the controller is introduced to not only receive sampled signals but also capture information on the duration time of each DoS attack. Note that the duration time of DoS attacks is usually both lower and upper bounded. Then the closed-loop system is modeled as an aperiodic sampled-data system closely related to both lower and upper bounds of duration time of DoS attacks. By introducing a novel looped functional, which caters for the N -order canonical Bessel–Legendre inequalities, some N -dependent stability criteria are presented for the resultant closed-loop system. It is worth pointing out that a number of identity formulas are uncovered, which enable us to apply the notable free-weighting matrix approach to derive less conservative stability criteria. A linear-matrix-inequality-based criterion is provided to design stabilizing state-feedback controllers against DoS attacks. A satellite control system is given to demonstrate the effectiveness of the proposed method.

Index Terms—Bessel–Legendre inequality, denial-of-service (DoS) attacks, looped functional, networked control systems (NCSs), resilient control.

I. INTRODUCTION

SINCE THE 1970s, automatic control has stepped into the era of networked control [1]–[3]. Boosted by advanced and rapid developments of communication technologies, networked control systems (NCSs) have made considerable progress and been found in abundance in

industrial applications, such as unmanned surface vehicles, unmanned space vehicles, smart grids, wastewater treatment processes, Internet-based teleoperation, and intelligent transportation systems [3]–[6]. The main reason is that constituent components in an NCS can be connected and communicated through a wired or wireless communication network, which brings several significant and important advantages, for example, increased capability for troubleshooting and maintenance, enhanced interchangeability and interoperability of devices, and improved reconfigurability of control systems [7]–[13]. Therefore, NCSs have attracted much attention from researchers and a great number of results have been reported on NCSs, see [14]–[20] and the references therein. However, due to the *shared* communication network, the safety of an NCS has become a critical issue. Malicious adversaries *intentionally* inject some attacks in a stealthy way into the *shared* communication network in order to make the NCS be disrupted completely [21]–[26]. In this situation, some existing control methodologies of NCSs are no longer effective when attacks are injected. Attacks in an NCS can be roughly categorized as [27]: 1) denial-of-service (DoS) attacks; 2) replay attacks; and 3) deception attacks [28]. DoS attacks aim to jam the communication network in an unpredictable time to prevent signals from being transmitted. A replay attack occurs when transmitted signals are fraudulently repeated to disrupt the routing protocols. Deception attacks may destroy the NCS by manipulating a number of false signals transmitted over the shared communication network. Under any kind of attack, the stability of an NCS, which is a fundamental issue, is threatened seriously [29], [30]. Therefore, it is of theoretical and practical significance to develop efficient schemes for NCSs to counter malicious attacks.

DoS attacks are regarded as one of the threats for modern communication networks [31]. A simple framework of an NCS under DoS attacks is shown in Fig. 1. Most DoS attacks are directed to deny signal transmission from the transmitter to the processor such that the NCS becomes unstable. In order to eliminate the effects of DoS attacks, a number of actions have been taken. To mention a few, a variety of potential attack scenarios by means of simulation is analyzed in [31]. By introducing a zero-sum game between the controller and the strategic jammers, optimal control is investigated even though a maximum number of jamming actions are taken over a prescribed control horizon by the jammers [32]. A general

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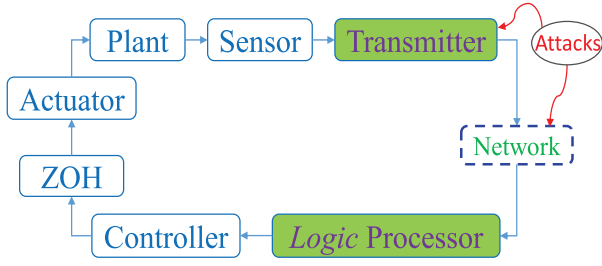


Fig. 1. Framework of an NCS under DoS attacks.

model for DoS attacks is established [34], in which an explicit characterization of DoS frequency and duration is disclosed in order that the closed-loop stability can be retained via *a priori* state feedback controller. An extension of this model can be found in [33] by introducing a family of impulsive controllers to maximize DoS frequency and duration such that the underlying closed-loop system may not be destabilized. Some other results on DoS attacks can be referred to [35]–[38].

This article is concerned with the resilient control design of a class of NCSs under DoS attacks. Different from some existing results, an effective approach to designing state feedback controllers against DoS attacks is proposed. Let the plant states be sampled periodically. DoS attacks jam the data communication from the transmitter to the controller in a stealthy way. By introducing a *logic* processor embedded in the controller, information on DoS attacks can be captured, by which DoS parameters, that is, frequency and the duration time of each DoS attack, can be calculated. As a result, the closed-loop system is modeled as an aperiodic sampled-data control system dependent closely on the maximum and minimum duration time of DoS attacks. By choosing a novel looped functional, which caters for the N -order affine canonical Bessel–Legendre inequalities, a couple of N -dependent stability criteria are derived by using a free-weighting matrix method, where a family of identity formulas are uncovered to characterize some certain intrinsic relations between those vectors inherited from the Bessel–Legendre inequality. An algorithm to design resilient controllers is presented in terms of several linear matrix inequalities with two tuning parameters. As long as the minimum and maximum duration times of DoS attacks are acquired, it is possible to design a stabilizing state feedback controller against DoS attacks, which is demonstrated through a satellite control system.

Notations: The notations throughout this article are standard. $\text{diag}\{\dots\}$ and $\text{col}\{\dots\}$ denote a block diagonal matrix and a block-column matrix (vector), respectively. $\begin{pmatrix} n \\ r \end{pmatrix} = n!/[r!(n-r)!]$. The symbol “ $\star$ ” in a symmetric block matrix stands for a term induced by symmetry. $\text{He}\{X\} = X + X^T$.

II. PROBLEM FORMULATION

The framework of an NCS is given in Fig. 1. In this section, we present a model for the NCS under DoS attacks.

A. Physical Plant, the Sensor, and the Transmitter

Consider the dynamics of the physical plant described by

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (1)$$

where $x \in \mathbb{R}^n$ is the system state, and $u \in \mathbb{R}^r$ is the control input; A and B are known real matrices with compatible dimensions. In this article, we suppose that the system state $x(t)$ is available for controller design, and (A, B) is stabilizable.

The sensor samples the system state with a period $T_0 > 0$. The sequence of sampling instants can be represented as $\{s_1, s_2, \dots, s_j, \dots\}$ with $s_j = jT_0$, $j \in \mathbb{N}$. The transmitter is event driven. Its aim is to encapsulate the current sampled state $x(s_j)$ with the timestamp s_j into a packet $(s_j, x(s_j))$, and transmit it to the processor through a communication network.

B. Network With DoS Attacks and the Logic Processor

Suppose that the network has a good quality of service so that the transmission delays can be neglectable. However, the network may be attacked by adversaries. There are kinds of ways for an adversary to attack the network. In this article, we focus on DoS attacks, under which the sampled-data packets are prevented from being transmitted by the transmitter and no packet is received by the processor. It is assumed that those packets that are not transmitted are no longer available. DoS attacks may be random or periodic, but the duration time is limited.

The logic processor is embedded in the controller. It consists of a buffer and a comparator. The buffer is used to store information about the newest packet received. Once the buffer is updated, the packet is immediately used to generate a new control signal by the controller. The sequence of timestamps of packets that the processor receives is denoted by $\{t_0, t_1, \dots, t_k, t_{k+1}, \dots\}$ with $t_0 = 0$. The comparator is taken to carry out some logical manipulations to capture information about DoS attacks. Its mechanism can be described concisely as follows.

The Comparator: Set $h_m = 100T_0$, $h_M = T_0$, $n_0 = 0$, $T_{\text{sum}} = 0$, and $\hat{t} = 0$. Suppose that a packet $(t_k, x(t_k))$ is received by the processor.

- 1) If $\Delta := t_k - \hat{t} > T_0$, then $T_{\text{sum}} = T_{\text{sum}} + \Delta - T_0$, $n_0 = n_0 + 1$.
- 2) If $\Delta > h_M$, then $h_M = \Delta$.
- 3) If $\Delta < h_m$, then $h_m = \Delta - T_0$.
- 4) $\hat{t} = t_k$.

In the above mechanism, n_0 records the total times of DoS attacks; h_m , h_M , and T_{sum} , respectively, are used to store the minimum duration time, the maximum duration time, and the sum of duration time of all DoS attacks during a certain time interval. If $\Delta > T_0$, which means that a DoS attack occurs,¹ then n_0 is increased by one, and the duration time is added to T_{sum} to store. Therefore, during $[0, T_f]$, where T_f is a certain time instant, the DoS frequency f_{DoS} and the average dwell time $\tilde{T}_{\text{DoS}}$ of each DoS attack can be calculated as

$$f_{\text{DoS}} = \frac{n_0}{T_f} \quad (2)$$

$$h_m \leq \frac{T_{\text{sum}}}{n_0} \leq \tilde{T}_{\text{DoS}} < \frac{T_{\text{sum}}}{n_0} + T_0 \leq h_M. \quad (3)$$

Take Fig. 2 as an example to show how the comparator works. At first, the packet $(T_0, x(T_0))$ is received by

¹If $\Delta \leq T_0$, such attacks are not malicious. In this case, we do not think an attack occurs.

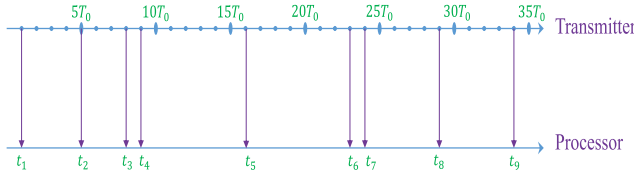


Fig. 2. Example to show how the comparator works.

the processor. Since the conditions 1)–3) are not satisfied, $\hat{t}$ changes its value with T_0 . Next, the packet $(5T_0, x(5T_0))$ is received. Since $\Delta = 5T_0 - T_0 = 4T_0 > T_0$, a DoS attack occurs during $(T_0, 5T_0)$, and its duration time belongs to $[3T_0, 4T_0)$. When the packet $(34T_0, x(34T_0))$ is received, since $\Delta = 34T_0 - 29T_0 = 5T_0 > T_0$, a DoS attack happens during $(29T_0, 34T_0)$. At this moment, the counter n_0 records 6 times attacks and $T_{\text{sum}} = 25T_0$. Hence, during $[0, 35T_0)$, the network has been attacked 6 times; the maximum (minimum) duration time is $7T_0$ ($2T_0$); and the average dwell time is bounded by $25T_0/6$ from below and by $(25/6 + 1)T_0$ from above. If setting the sampling period T_0 to be 0.1 s, then the DoS frequency during $[0, 35T_0)$ is 1.71 times per second and the average dwell time is between 0.42 s and 0.52 s.

From above, it is clear to see that the comparator can provide some important information about DoS attacks, including DoS frequency and average dwell time of each DoS attack. From (2) and (3), we have

$$h_m \leq \frac{T_{\text{sum}}/T_f}{f_{\text{DoS}}} \leq \tilde{T}_{\text{DoS}} < \frac{T_{\text{sum}}/T_f}{f_{\text{DoS}}} + T_0 \leq h_M \quad (4)$$

which follows:

$$\frac{T_{\text{sum}}/T_f}{h_M} < f_{\text{DoS}} \leq \frac{T_{\text{sum}}/T_f}{h_m}. \quad (5)$$

Thus, both DoS frequency and DoS average dwell time are heavily dependent on the minimum and maximum duration time h_m and h_M .

C. Controller and the Resultant Closed-Loop System

The controller collaborates with the logic processor closely. Once the processor receives a new packet, the controller uses it immediately to generate a new control signal to be held by the zero-order hold (ZOH) so that the physical plant can be actuated. The control signal is calculated by

$$u(t_k) = Fx(t_k), \quad k \in \mathcal{I} \quad (6)$$

where F is the control gain, and $\mathcal{I} = \{0, 1, 2, \dots\}$ is the index set. By the property of ZOH, the control input can be given by

$$u(t) = Fx(t_k) \quad t \in [t_k, t_{k+1}), \quad k \in \mathcal{I}. \quad (7)$$

From the mechanism of the comparator, it is clear that

$$\Delta_k \triangleq t_{k+1} - t_k \in [h_m, h_M], \quad k \in \mathcal{I}. \quad (8)$$

Substituting (7) into (1), the closed-loop system is obtained as

$$\dot{x}(t) = Ax(t) + BFx(t_k), \quad t \in [t_k, t_{k+1}). \quad (9)$$

From the above analysis, one can see clearly that, the states of the physical plant are sampled *periodically*, while under

DoS attacks, the NCS is modeled as an *aperiodic* sampled-data control system. However, this model is *not* suitable for replay attacks and deception attacks. The problem to be addressed in this article is given as follows.

Resilient Controller Design: For a given sampling period T_0 , based on the values of h_m and h_M that the logic processor provides, design a resilient controller (7) against DoS attacks. That is, determine the controller gain F such that the closed-loop system is asymptotically stable.

Remark 1: Recalling some existing results on DoS attacks, the objective is to design a *sampling logic* in [34] or a *proper sampling period* T_0 in [33] for a *given* stabilizing controller such that the stabilizing controller is robust to DoS attacks. However, this article aims to present an algorithm to *design stabilizing controllers* against DoS attacks for a *given* sampling period T_0 . Therefore, this article paves a different way to study NCSs subject to DoS attacks.

In the next sections, we are to address the above problem in two steps. First, we establish some general stability criteria for the system (9) using a looped-functional method [39] plus N -order canonical Bessel–Legendre inequalities [41]. Second, we propose a method to design suitable control gain F based on the obtained stability criteria.

III. STABILITY CRITERIA BASED ON N -ORDER CANONICAL BESSEL–LEGENDRE INEQUALITIES

In this section, we present some novel stability criteria for (9) based on N -order canonical Bessel–Legendre inequalities using a looped-functional method. In doing so, we first introduce two N -order canonical Bessel–Legendre inequalities, which are originally proposed in [40] and adapted in [41] in a simple form that is easy to use.

A. N -Order Canonical Bessel–Legendre Inequalities

Lemma 1: For an integer $N \geq 0$, a real matrix $R > 0$, two scalars a and b with $b > a$, and a vector-valued differentiable function $\omega : [a, b] \rightarrow \mathbb{R}^n$ such that the integrations below are well defined, then

$$-\int_a^b \dot{\omega}^T(s) R \dot{\omega}(s) ds \leq -\frac{1}{b-a} \mathcal{X}_N^T \Pi_N^T \mathcal{R}_N \Pi_N \mathcal{X}_N \quad (10)$$

$$= -\frac{1}{b-a} \mathcal{Y}_N^T \Pi_N^T \mathcal{R}_N \Pi_N \mathcal{Y}_N \quad (11)$$

where $\mathcal{R}_N := \text{diag}\{R, 3R, \dots, (2N+1)R\}$ and $\Pi_N := \Theta_N \Lambda_N$ with

$$\Theta_N := \begin{bmatrix} I & 0 & \cdots & 0 \\ I & (-1)^1 \binom{1}{1} \binom{2}{1} I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ I & (-1)^1 \binom{N}{1} \binom{N+1}{1} I & \cdots & (-1)^N \binom{N}{N} \binom{2N}{N} I \end{bmatrix}$$

$$\Lambda_N := \begin{bmatrix} I & -I & 0 & 0 & \cdots & 0 \\ 0 & -I & I & 0 & \cdots & 0 \\ 0 & -I & 0 & 2I & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & -I & 0 & 0 & \cdots & NI \end{bmatrix}$$

$$\mathcal{X}_N := \text{col}\{\omega(b), \omega(a), \gamma_1, \dots, \gamma_N\}, \gamma_j := \int_a^b \frac{(b-s)^{j-1}}{(b-a)^j} \omega(s) ds$$

$$\mathcal{Y}_N := \text{col}\{\omega(a), \omega(b), \tilde{\gamma}_1, \dots, \tilde{\gamma}_N\}, \tilde{\gamma}_j := \int_a^b \frac{(s-a)^{j-1}}{(b-a)^j} \omega(s) ds.$$

Let ϖ be a p -dimensional vector satisfying $\mathcal{X}_N = \mathcal{C}\varpi$ or $\mathcal{Y}_N = \mathcal{C}\varpi$, where $\mathcal{C}$ is a certain real matrix. Then the above inequalities (10) and (11) can be of the following general form:

$$-\int_a^b \dot{\omega}^T(s) R \dot{\omega}(s) ds \leq \varpi^T \mathcal{F}_a^b(\mathcal{C}, R, \mathcal{M}) \varpi \quad (12)$$

$$\mathcal{F}_a^b(\mathcal{C}, R, \mathcal{M}) := \mathcal{C}^T \Pi_N^T \mathcal{M} + \mathcal{M}^T \Pi_N \mathcal{C} + (b-a) \mathcal{M}^T \mathcal{R}_N^{-1} \mathcal{M} \quad (13)$$

where $\mathcal{M}$ is a free-weighting real matrix.

An advantage of the inequality (12) is that the dimension of the vector ϖ is adjustable, either larger or smaller than the dimension of $\mathcal{X}_N$ or $\mathcal{Y}_N$. A large dimension of ϖ leads to a large weighting matrix $\mathcal{M}$.

As an application of (12), we denote

$$\rho(t) := \text{col}\{\rho_1(t), \rho_2(t), \dots, \rho_N(t)\} \quad (14)$$

$$\wp(t) := \text{col}\{\wp_1(t), \wp_2(t), \dots, \wp_N(t)\} \quad (15)$$

where

$$\begin{cases} \rho_i(t) := \int_{t_k}^t \frac{(t-s)^{i-1}}{(t-t_k)^i} x(s) ds \\ \wp_i(t) := \int_t^{t_{k+1}} \frac{(s-t)^{i-1}}{(t_{k+1}-t)^i} x(s) ds \end{cases} \quad (16)$$

with $i = 1, 2, \dots, N$, and introduce a vector as

$$\xi(t) := \text{col}\{x(t), x(t_k), x(t_{k+1}), (t-t_k)\rho(t), (t_{k+1}-t)\wp(t), \rho(t), \wp(t)\}. \quad (17)$$

Applying the inequality (12) to $-\int_{t_k}^t \dot{x}^T(s) R_1 \dot{x}(s) ds$ and $-\int_t^{t_{k+1}} \dot{x}^T(s) R_2 \dot{x}(s) ds$ yields, respectively

$$-\int_{t_k}^t \dot{x}^T(s) R_1 \dot{x}(s) ds \leq \xi^T(t) \mathcal{F}_{t_k}^t(\mathcal{C}_1, R_1, \mathcal{M}_1) \xi(t) \quad (18)$$

$$-\int_t^{t_{k+1}} \dot{x}^T(s) R_2 \dot{x}(s) ds \leq \xi^T(t) \mathcal{F}_t^{t_{k+1}}(\mathcal{C}_2, R_2, \mathcal{M}_2) \xi(t) \quad (19)$$

where $R_1 > 0$, $R_2 > 0$, $\mathcal{M}_1$ and $\mathcal{M}_2$ are real matrices with appropriate dimensions; and

$$\mathcal{C}_1 := \text{col}\{e_1, e_2, e_6\}, \quad \mathcal{C}_2 := \text{col}\{e_1, e_3, e_7\} \quad (20)$$

where $e_1, e_2, \dots, e_7$ are row-block vectors taking from an identity matrix such that $x(t) = e_1 \xi(t)$, $x(t_k) = e_2 \xi(t)$, $\dots$, $\wp(t) = e_7 \xi(t)$. The inequalities (18) and (19) are useful in the stability analysis of the system (9), which can be seen in the next sections.

B. Some New Identity Formulas

The above canonical Bessel–Legendre inequalities introduce some extra vectors $\rho_i(t)$ and $\wp_i(t)$ in (16), which are not convenient to use in the stability analysis of the system (9). The following identity formulas establish bridges between them and the system states $x(t)$, $x(t_k)$, and $x(t_{k+1})$.

From the system (9), for $i = 1, 2, \dots, N-1$, the following hold for $t \in [t_k, t_{k+1}]$:

$$\begin{aligned} \int_{t_k}^t \dot{x}(s) ds &= \int_{t_k}^t [Ax(s) + Bx(t_k)] ds \\ \int_t^{t_{k+1}} \dot{x}(s) ds &= \int_t^{t_{k+1}} [Ax(s) + Bx(t_k)] ds \\ \int_{t_k}^t (t-s)^i \dot{x}(s) ds &= \int_{t_k}^t (t-s)^i [Ax(s) + Bx(t_k)] ds \\ \int_t^{t_{k+1}} (s-t)^i \dot{x}(s) ds &= \int_t^{t_{k+1}} (s-t)^i [Ax(s) + Bx(t_k)] ds. \end{aligned}$$

Simple calculation gives

$$\begin{cases} x(t) - x(t_k) = (t-t_k)[A\rho_1(t) + Bx(t_k)] \\ x(t_{k+1}) - x(t) = (t_{k+1}-t)[A\wp_1(t) + Bx(t_k)] \\ -x(t_k) + i\rho_i(t) = (t-t_k)[A\rho_{i+1}(t) + \frac{1}{i+1}Bx(t_k)] \\ x(t_{k+1}) - i\wp_i(t) = (t_{k+1}-t)[A\wp_{i+1}(t) + \frac{1}{i+1}Bx(t_k)] \end{cases} \quad (21)$$

where $i = 1, 2, \dots, N-1$. These identity formulas can be expressed in a compact form as

$$\begin{cases} \tilde{\Lambda}_{N-1} \mathcal{C}_1 \xi(t) = (t-t_k) \Xi_N \mathcal{C}_3 \xi(t) \\ -\tilde{\Lambda}_{N-1} \mathcal{C}_2 \xi(t) = (t_{k+1}-t) \Xi_N \mathcal{C}_4 \xi(t) \end{cases} \quad (22)$$

where $\tilde{\Lambda}_{N-1} = [\Lambda_{N-1} \ 0]$, $\mathcal{C}_3 = \text{col}\{e_2, e_6\}$, $\mathcal{C}_4 = \text{col}\{e_2, e_7\}$, and

$$\Xi_N := \begin{bmatrix} BF & A & 0 & \cdots & 0 \\ \frac{1}{2}BF & 0 & A & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{N}BF & 0 & 0 & \cdots & A \end{bmatrix}. \quad (23)$$

Remark 2: The identity formulas in (22) are derived from the system (9), which provide a family of identity formulas to characterize the intrinsic relations between the vectors $\rho_i(t)$ and $\wp_i(t)$, and the state vectors $x(t)$, $x(t_k)$, and $x(t_{k+1})$. These identity formulas enable us to employ the free-weighting matrix method to obtain some less conservative stability criteria for the system (9).

C. Stability Criteria

A looped functional candidate is constructed as

$$V(x) = x^T(t)Px(t) + V_1(x) + V_2(x) + V_3(x) + V_4(x) \quad (24)$$

where $P > 0$ and

$$\begin{aligned} V_1(x) &= 2\eta^T(t)Q_N \begin{bmatrix} (t_{k+1}-t)[x(t) - x(t_k)] \\ (t-t_k)[x(t) - x(t_{k+1})] \end{bmatrix} \\ V_2(x) &= 2[x(t) - x(t_k)]^T S_1 [x(t) - x(t_{k+1})] \\ V_3(x) &= (t_{k+1}-t)(t-t_k) \begin{bmatrix} x(t_k) \\ x(t_{k+1}) \end{bmatrix}^T S_2 \begin{bmatrix} x(t_k) \\ x(t_{k+1}) \end{bmatrix} \\ V_4(x) &= (t_{k+1}-t) \int_{t_k}^t \dot{x}^T(s) R_1 \dot{x}(s) ds \\ &\quad - (t-t_k) \int_t^{t_{k+1}} \dot{x}^T(s) R_2 \dot{x}(s) ds \end{aligned}$$

with $R_1 > 0, R_2 > 0, S_2 = S_2^T$, and S_1, Q_N being compatible real matrices to be determined; and

$$\eta(t) := \text{col}\{x(t), x(t_k), x(t_{k+1}), (t - t_k)\rho(t), (t_{k+1} - t)\varrho(t)\}.$$

Notice that $V(x(t)) = x^T(t)Px(t)$ at $t = t_k$ ($k = 1, 2, \dots$). Thus, $V(x)$ is a looped functional [42], [43]. Another feature of $V(x)$ is that a general vector $\eta(t)$ is introduced in $V_1(x)$, which caters for the use of the canonical Bessel–Legendre inequalities in Lemma 1 or in (12). In the following, we state and establish two stability criteria based on (24).

1) Stability Criterion I:

Proposition 1: For given two constants h_m and h_M with $h_M \geq h_m > 0$ and a positive integer N , the system (9) is asymptotically stable if there exist compatible real matrices $P > 0, R_1 > 0, R_2 > 0, Q_N, S_1, S_2 = S_2^T, \mathcal{M}_1, \mathcal{M}_2$, and L_i ($i = 1, 2, 3, 4$) such that, for $\Delta_k \in \{h_m, h_M\}$

$$\begin{bmatrix} \Phi_0 + \Delta_k \Phi_2 & \Delta_k \mathcal{M}_1^T \\ \Delta_k \mathcal{M}_1 & -\Delta_k \mathcal{R}_{1N} \end{bmatrix} < 0 \quad (25)$$

$$\begin{bmatrix} \Phi_0 + \Delta_k \Phi_1 & \Delta_k \mathcal{M}_2^T \\ \Delta_k \mathcal{M}_2 & -\Delta_k \mathcal{R}_{2N} \end{bmatrix} < 0 \quad (26)$$

where $\mathcal{R}_{iN} := \text{diag}\{R_i, 3R_i, \dots, (2N+1)R_i\}$ ($i = 1, 2$), and

$$\begin{aligned} \Phi_0 &:= \text{He}\{e_1^T P C_0 + C_1^T Q_N C_4 + C_0^T S_1 E_2 C_4 - C_0^T S_1^T E_1 C_4\} \\ &+ \text{He}\{e_1^T \Pi_N^T \mathcal{M}_1 + e_2^T \Pi_N^T \mathcal{M}_2 + L_1 \tilde{\Lambda}_{N-1} \mathcal{C}_1\} \\ &+ \text{He}\{L_2 \tilde{\Lambda}_{N-1} \mathcal{C}_2 + L_3 e_4 + L_4 e_5\} \end{aligned} \quad (27)$$

$$\begin{aligned} \Phi_1 &:= \text{He}\{C_1^T Q_N E_1^T C_0 - C_2^T Q_N E_1^T E_1 C_4 + L_2 \Xi_N \mathcal{C}_4\} \\ &- \text{He}\{L_4 e_7\} + C_3^T S_2 C_3 + C_0^T R_1 C_0 \end{aligned} \quad (28)$$

$$\begin{aligned} \Phi_2 &:= \text{He}\{C_1^T Q_N E_2^T C_0 + C_2^T Q_N E_2^T E_2 C_4 - L_1 \Xi_N \mathcal{C}_3\} \\ &- \text{He}\{L_3 e_6\} - C_3^T S_2 C_3 + C_0^T R_2 C_0 \end{aligned} \quad (29)$$

with $E_1 = [I \ 0]$, $E_2 = [0 \ I]$, and

$$C_0 := A e_1 + B F e_2, \quad C_1 := \text{col}\{e_1, e_2, e_3, e_4, e_5\}$$

$$C_2 := \text{col}\{C_0, 0, 0, e_1, \Gamma_N e_6, -e_1, -\Gamma_N e_7\}$$

$$\Gamma_N := \begin{bmatrix} I & -I & 0 & \cdots & 0 & 0 \\ 0 & 2I & -2I & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ddots & -(N-2)I & 0 \\ 0 & 0 & 0 & \cdots & (N-1)I & -(N-1)I \end{bmatrix}$$

$$C_3 := \text{col}\{e_2, e_3\}, \quad C_4 := \text{col}\{e_2 - e_1, e_1 - e_3\}.$$

Proof: Taking the time derivative of $V(x)$ yields

$$\dot{V}(x) = 2x^T(t)P\dot{x}(t) + \dot{V}_1(x) + \dot{V}_2(x) + \dot{V}_3(x) + \dot{V}_4(x) \quad (30)$$

where

$$\begin{aligned} \dot{V}_1(x) &= 2\eta^T(t)Q_N \begin{bmatrix} (t_{k+1} - t)\dot{x}(t) - [x(t) - x(t_k)] \\ (t - t_k)\dot{x}(t) + [x(t) - x(t_{k+1})] \end{bmatrix} \\ &+ 2\dot{\eta}^T(t)Q_N \begin{bmatrix} (t_{k+1} - t)[x(t) - x(t_k)] \\ (t - t_k)[x(t) - x(t_{k+1})] \end{bmatrix} \end{aligned}$$

$$\dot{V}_2(x) = 2\dot{x}^T(t)S_1[x(t) - x(t_{k+1})] + 2[x(t) - x(t_k)]^T S_1 \dot{x}(t)$$

$$\dot{V}_3(x) = [(t_{k+1} - t) - (t - t_k)] \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix}^T S_2 \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix}$$

$$\dot{V}_4(x) = \dot{x}^T(t)[(t_{k+1} - t)R_1 + (t - t_k)R_2]\dot{x}(t)$$

$$\begin{aligned} &- \int_{t_k}^t \dot{x}^T(s)R_1 \dot{x}(s)ds - \int_t^{t_{k+1}} \dot{x}^T(s)R_2 \dot{x}(s)ds \\ &\leq \dot{x}^T(t)[(t_{k+1} - t)R_1 + (t - t_k)R_2]\dot{x}(t) \\ &+ \xi^T(t) \left[\mathcal{F}_{t_k}^t(\mathcal{C}_1, R_1, \mathcal{M}_1) + \mathcal{F}_t^{t_{k+1}}(\mathcal{C}_2, R_2, \mathcal{M}_2) \right] \xi(t) \end{aligned}$$

where $\mathcal{F}_a^b(\mathcal{C}, R, \mathcal{M})$ is defined in (13). Notice that

$$\dot{x}(t) = C_0 \xi(t), \quad \eta(t) = C_1 \xi(t), \quad \dot{\eta}(t) = C_2 \xi(t) \quad (31)$$

where C_0, C_1 , and C_2 are defined in Proposition 1. After some simple algebraic manipulations, one has

$$\begin{aligned} \dot{V}_1(x) &= 2\xi^T(t)C_1^T Q_N \{[(t_{k+1} - t)E_1^T + (t - t_k)E_2^T]C_0 \\ &+ C_4\} \xi(t) \\ &+ 2\xi^T(t)C_2^T Q_N [(t - t_k)E_2^T E_2 C_4 \\ &- (t_{k+1} - t)E_1^T E_1 C_4] \xi(t) \\ \dot{V}_2(x) &= 2\xi^T(t)C_0^T S_1 E_2 C_4 \xi(t) - 2\xi^T(t)C_0^T S_1^T E_1 C_4 \xi(t) \\ \dot{V}_3(x) &= [(t_{k+1} - t) - (t - t_k)] \xi^T(t)C_3^T S_2 C_3 \xi(t) \end{aligned}$$

where C_3 and C_4 are given in Proposition 1.

On the other hand, note that

$$e_4 \xi(t) = (t - t_k)e_6 \xi(t), \quad e_5 \xi(t) = (t_{k+1} - t)e_7 \xi(t). \quad (32)$$

Then from (22) and (32), for any compatible real matrices L_1, L_2, L_3 , and L_4 , the following “zero” equations hold:

$$0 = 2\xi^T(t)L_1 [\tilde{\Lambda}_{N-1} \mathcal{C}_1 - (t - t_k)\Xi_N \mathcal{C}_3] \xi(t) \quad (33)$$

$$0 = 2\xi^T(t)L_2 [\tilde{\Lambda}_{N-1} \mathcal{C}_2 + (t_{k+1} - t)\Xi_N \mathcal{C}_4] \xi(t) \quad (34)$$

$$0 = 2\xi^T(t)L_3 [e_4 - (t - t_k)e_6] \xi(t) \quad (35)$$

$$0 = 2\xi^T(t)L_4 [e_5 - (t_{k+1} - t)e_7] \xi(t). \quad (36)$$

Based on (30), and (33)–(36), after rearrangement, we have that

$$\begin{aligned} \dot{V}(x) &\leq \xi^T(t) \left[\Phi_0 + (t_{k+1} - t)\tilde{\Phi}_1 + (t - t_k)\tilde{\Phi}_2 \right] \xi(t) \\ &= \xi^T(t) \left[\frac{t_{k+1} - t}{\Delta_k} (\Phi_0 + \Delta_k \tilde{\Phi}_1) + \frac{t - t_k}{\Delta_k} (\Phi_0 + \Delta_k \tilde{\Phi}_2) \right] \xi(t) \end{aligned}$$

where Φ_0 is defined in (27), and

$$\tilde{\Phi}_1 := \Phi_1 + \mathcal{M}_2^T \mathcal{R}_{2N}^{-1} \mathcal{M}_2, \quad \tilde{\Phi}_2 := \Phi_2 + \mathcal{M}_1^T \mathcal{R}_{1N}^{-1} \mathcal{M}_1 \quad (37)$$

with Φ_1 and Φ_2 given in (28) and (29), respectively.

By [39, Th. 1], a sufficient condition on the stability of (9) is that

$$\Phi_0 + \Delta_k \tilde{\Phi}_2 < 0, \quad \Phi_0 + \Delta_k \tilde{\Phi}_1 < 0 \quad (38)$$

which are equivalent to (25) and (26), respectively. ■

Remark 3: Proposition 1 delivers an N -dependent stability criterion for the system (9). The significance lies in that:

- 1) the looped functional is introduced to cater for the use of the Bessel–Legendre inequality;
- 2) the vectors $(t - t_k)\rho(t)$ and $(t_{k+1} - t)\varrho(t)$ are introduced in $\xi(t)$ to avoid the appearance of $(t - t_k)^2$ and $(t_{k+1} - t)^2$ in the derivative of $V(x)$;
- 3) the free-weighting matrix approach is developed through some novel “zero” equations in (33)–(36).

TABLE I
MAXIMUM ADMISSIBLE UPPER BOUND h_M FOR $h_m = 0.001$
FOR THE SYSTEM (9) WITH (39)

Method	h_M
[39]	2.62
[42]	3.2632
Proposition 1 ($N = 2$)	3.267
Proposition 1 ($N = 3$)	3.2714
Proposition 2 ($N = 2$)	3.2636
Proposition 2 ($N = 3$)	3.2712
Analytic bound	3.2722

To show the effectiveness of Proposition 1, we now consider the system (9) with

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -0.9 \end{bmatrix}, \quad BF = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix}. \quad (39)$$

This system is investigated in [39] and [42], where h_m and h_M are regarded as the lower and upper bounds of time distance between any two consecutive samplings. For comparison, we calculate the allowable maximum upper bound of h_M for a small $h_m = 0.001$. The results are listed in Table I, from which one can see clearly that Proposition 1 outperforms the ones in [39] and [42]. On the other hand, for periodic samplings, that is, $h_m = h_M = h_0$, it is found that the system is stable for $h_0 \leq 3.2722$, which further illustrates the effectiveness of Proposition 1.

2) *Stability Criterion II*: Although Proposition 1 presents a good stability criterion for the closed-loop system (9), it is challenging to design a suitable control gain F based on Proposition 1. In fact, in Proposition 1, the matrix F is coupled with the Lyapunov matrix P as well as Q_N, S_1, R_1, R_2, L_1 , and L_2 . Such couplings definitely are an impediment to solve out the control gain F . In the following, we provide an alternative stability criterion, from which the above couplings are removed.

After an insightful observation of Proposition 1, it is found that the above couplings occur in two situations. The first situation is when $\dot{x}(t)$ is replaced with $C_0\xi(t)$. For example, when $2x^T(t)P\dot{x}(t)$ is replaced with $2x^T(t)PC_0\xi(t)$, the gain matrix F is coupled with P . The other situation is when the “zero” equations (33) and (34) are introduced. Since the gain matrix F is sitting in the matrix Ξ_N , it is coupled with L_1 and L_2 when (33) and (34) are used.

In order not to couple F with Q_N, S_1, R_1 , and R_2 , an easy way is to keep $\dot{x}(t)$ unchangeable rather than replace it with $C_0\xi(t)$. However, it is not so easy to decouple F with L_1 and L_2 unless F does not dwell in the matrix Ξ_N . In the sequel, we provide a method to remove the matrix F from the matrix Ξ_N .

From (21), we have

$$\begin{cases} x(t) - x(t_k) = (t - t_k)[A\rho_1(t) + BFx(t_k)] \\ -2x(t_k) + 2\rho_1(t) = (t - t_k)[2A\rho_2(t) + BFx(t_k)] \\ \vdots \\ -(N-1)x(t_k) + (N-1)(N-2)\rho_{N-2}(t) \\ = (t - t_k)[(N-1)A\rho_{N-1}(t) + BFx(t_k)] \\ -Nx(t_k) + N(N-1)\rho_{N-1}(t) \\ = (t - t_k)[NA\rho_N(t) + BFx(t_k)]. \end{cases}$$

Performing a number of subtractions, (i.e., the i th equation subtracts the $(i-1)$ th equation, $i = N, N-1, \dots, 2$) yields

$$\begin{cases} -x(t) - x(t_k) + 2\rho_1(t) = (t - t_k)A[2\rho_2(t) - \rho_1(t)] \\ -x(t_k) + 6\rho_2(t) - 2\rho_1(t) = (t - t_k)A[3\rho_3(t) - 2\rho_2(t)] \\ \vdots \\ -x(t_k) + N(N-1)\rho_{N-1}(t) - (N-1)(N-2)\rho_{N-2}(t) \\ = (t - t_k)A[N\rho_N(t) - (N-1)\rho_{N-1}(t)] \end{cases}$$

which follows a compact form as:

$$\Sigma_N \mathcal{C}_1 \xi(t) = \tilde{\Xi}_N e_4 \xi(t) \quad (40)$$

where $N \geq 2$ and

$$\Sigma_N = \begin{bmatrix} -I & -I & 2I & 0 & \cdots & 0 & 0 & 0 \\ 0 & -I & -a_2 I & a_3 I & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & -I & 0 & 0 & \cdots & -a_{N-1} I & a_N I & 0 \end{bmatrix}$$

$$\tilde{\Xi}_N = \begin{bmatrix} -A & 2A & 0 & \cdots & 0 & 0 \\ 0 & -2A & 3A & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -(N-1)A & NA \end{bmatrix}$$

where $a_i = i(i-1)$ ($i = 2, 3, \dots, N$). Similarly, one has

$$-\Sigma_N \mathcal{C}_2 \xi(t) = \tilde{\Xi}_N e_5 \xi(t). \quad (41)$$

Remark 4: It is clear to see that, based on (21), two new identity formulas (40) and (41) are obtained, where the matrix $\tilde{\Xi}_N$ is independent of the gain matrix F . This feature allows the gain matrix F not to be coupled with L_1 and L_2 as done in Proposition 1.

Now we state the second stability criterion for the system (9).

Proposition 2: For given two constants h_m and h_M with $h_M \geq h_m > 0$ and a positive integer N , the system (9) is asymptotically stable if there exist compatible real matrices $P > 0$, $R_1 > 0$, $R_2 > 0$, $Q_N, S_1, S_2 = S_2^T, \mathcal{M}_1, \mathcal{M}_2$, L_i ($i = 1, 2, \dots, 8$), Y_1 , and Y_2 such that, for $\Delta_k \in \{h_m, h_M\}$

$$\Upsilon_1 := \begin{bmatrix} \text{He}\{Y_1\} + \Delta_k R_1 & \Psi_{01} + \Psi_{02} + \Delta_k \Psi_{11} & 0 \\ \star & \Psi_{03} + \Psi_{04} + \Delta_k \Psi_{12} & \Delta_k \mathcal{M}_2^T \\ \star & \star & -\Delta_k \mathcal{R}_{2N} \end{bmatrix} < 0 \quad (42)$$

$$\Upsilon_2 := \begin{bmatrix} \text{He}\{Y_1\} + \Delta_k R_2 & \Psi_{01} + \Psi_{02} + \Delta_k \Psi_{21} & 0 \\ \star & \Psi_{03} + \Psi_{04} + \Delta_k \Psi_{22} & \Delta_k \mathcal{M}_1^T \\ \star & \star & -\Delta_k \mathcal{R}_{1N} \end{bmatrix} < 0 \quad (43)$$

where

$$\begin{aligned} \Psi_{01} &:= Pe_1 + (S_1 E_2 - S_1^T E_1)C_4 - Y_1 C_0 + Y_2^T \\ \Psi_{02} &:= L_1 e_4 + L_2 e_5 + L_3 [\Sigma_N \mathcal{C}_1 - \tilde{\Xi}_N e_4] \\ &\quad + L_4 [\Sigma_N \mathcal{C}_2 + \tilde{\Xi}_N e_5] \\ \Psi_{03} &:= \text{He}\{C_1^T Q_N C_4 + \mathcal{C}_1^T \Pi_N^T \mathcal{M}_1 + \mathcal{C}_2^T \Pi_N^T \mathcal{M}_2 - Y_2 C_0\} \\ \Psi_{04} &:= \text{He}\{L_7 (\Sigma_N \mathcal{C}_1 - \tilde{\Xi}_N e_4) + L_8 (\Sigma_N \mathcal{C}_2 + \tilde{\Xi}_N e_5)\} \\ &\quad + \text{He}\{L_5 e_4 + L_6 e_5\} \end{aligned}$$

$$\begin{aligned}
\Psi_{11} &:= E_1 Q_N^T C_1 - C_{21}^T Q_N E_1^T E_1 C_4 - L_2 e_7 \\
\Psi_{12} &:= C_3^T S_2 C_3 - \text{He}\{C_{22}^T Q_N E_1^T E_1 C_4 + L_6 e_7\} \\
\Psi_{21} &:= E_2 Q_N^T C_1 + C_{21}^T Q_N E_2^T E_2 C_4 - L_1 e_6 \\
\Psi_{22} &:= \text{He}\{C_{22}^T Q_N E_2^T E_2 C_4 - L_5 e_6\} - C_3^T S_2 C_3 \\
C_{21} &:= \text{col}\{I, 0, 0, 0, 0, 0, 0\} \\
C_{22} &:= \text{col}\{0, 0, 0, e_1, \Gamma_N e_6, -e_1, -\Gamma_N e_7\}
\end{aligned}$$

and the other notations are defined in Proposition 1.

Proof: In the proof of Proposition 1, we keep $\dot{x}(t)$ unchanged. In this situation, $\dot{\eta}(t)$ can be described as

$$\dot{\eta}(t) = C_{21}\dot{x}(t) + C_{22}\xi(t). \quad (44)$$

Then similar to the proof in Proposition 1, we have

$$\dot{V}(x) \leq \begin{bmatrix} \dot{x}(t) \\ \xi(t) \end{bmatrix}^T \left[\tilde{\Omega}_0 + (t_{k+1} - t)\tilde{\Omega}_1 + (t - t_k)\tilde{\Omega}_2 \right] \begin{bmatrix} \dot{x}(t) \\ \xi(t) \end{bmatrix} \quad (45)$$

where

$$\begin{aligned}
\tilde{\Omega}_0 &:= \begin{bmatrix} 0 & Pe_1 + (S_1 E_2 - S_1^T E_1) C_4 \\ \star & \text{He}\{C_1^T Q_N C_4 + \mathcal{C}_1^T \Pi_N^T \mathcal{M}_1 + \mathcal{C}_2^T \Pi_N^T \mathcal{M}_2\} \end{bmatrix} \\
\tilde{\Omega}_1 &:= \begin{bmatrix} R_1 & E_1 Q_N^T C_1 - C_{21}^T Q_N E_1^T E_1 C_4 \\ \star & C_3^T S_2 C_3 - \text{He}\{C_{22}^T Q_N E_1^T E_1 C_4\} + \mathcal{M}_2^T \mathcal{R}_{2N}^{-1} \mathcal{M}_2 \end{bmatrix} \\
\tilde{\Omega}_2 &:= \begin{bmatrix} R_2 & E_2 Q_N^T C_1 + C_{21}^T Q_N E_2^T E_2 C_4 \\ \star & \text{He}\{C_{22}^T Q_N E_2^T E_2 C_4\} - C_3^T S_2 C_3 + \mathcal{M}_1^T \mathcal{R}_{1N}^{-1} \mathcal{M}_1 \end{bmatrix}.
\end{aligned}$$

Utilizing the identity equations in (32), (40), and (41), for any compatible real matrices L_i ($i = 1, 2, \dots, 8$), one obtains

$$\begin{cases} 0 = 2[\dot{x}^T(t)L_1 + \xi^T(t)L_5][e_4 - (t - t_k)e_6]\xi(t) \\ 0 = 2[\dot{x}^T(t)L_2 + \xi^T(t)L_6][e_5 - (t_{k+1} - t)e_7]\xi(t) \\ 0 = 2[\dot{x}^T(t)L_3 + \xi^T(t)L_7][\Sigma_N \mathcal{C}_1 - \tilde{\Xi}_N e_4]\xi(t) \\ 0 = 2[\dot{x}^T(t)L_4 + \xi^T(t)L_8][\Sigma_N \mathcal{C}_2 + \tilde{\Xi}_N e_5]\xi(t). \end{cases} \quad (46)$$

On the other hand, for any real matrices $Y_j \in \mathbb{R}^{n \times n}$ ($j = 1, 2$)

$$0 = 2[\dot{x}^T(t)Y_1 + \xi^T(t)Y_2][\dot{x}(t) - Ax(t) - BFx(t_k)]$$

which leads to

$$0 = 2[\dot{x}^T(t)Y_1 + \xi^T(t)Y_2][\dot{x}(t) - C_0 \xi(t)]. \quad (47)$$

Adding the “zero” terms in (46) and (47) into (45) yields

$$\dot{V}(x) \leq \begin{bmatrix} \dot{x}(t) \\ \xi(t) \end{bmatrix}^T \left[\Omega_0 + (t_{k+1} - t)\Omega_1 + (t - t_k)\Omega_2 \right] \begin{bmatrix} \dot{x}(t) \\ \xi(t) \end{bmatrix}$$

where

$$\begin{aligned}
\Omega_0 &:= \begin{bmatrix} \text{He}\{Y_1\} & \Psi_{01} + \Psi_{02} \\ \star & \Psi_{03} + \Psi_{04} \end{bmatrix} \\
\Omega_1 &:= \begin{bmatrix} R_1 & \Psi_{11} \\ \star & \Psi_{12} + \mathcal{M}_2^T \mathcal{R}_{2N}^{-1} \mathcal{M}_2 \end{bmatrix} \\
\Omega_2 &:= \begin{bmatrix} R_2 & \Psi_{21} \\ \star & \Psi_{22} + \mathcal{M}_1^T \mathcal{R}_{1N}^{-1} \mathcal{M}_1 \end{bmatrix}.
\end{aligned}$$

If the matrix inequalities (42) and (43) are satisfied, by the Schur complement, we have $\Omega_0 + \Delta_k \Omega_1 < 0$ and $\Omega_0 + \Delta_k \Omega_2 < 0$, which means that $\dot{V}(x) < 0$. The conclusion can thus be drawn using [39, Th. 1]. ■

Remark 5: Proposition 2 presents a different stability condition for the system (9) based on the identity formulas (40) and (41), where the matrix $\tilde{\Xi}_N$ does not include the gain F . Now, we make a comparison with Proposition 1 based on the system (9) with (39). The calculation results are also given in Table I. From this table, one can see that the obtained maximum h_M is almost the same as that obtained by Proposition 1 with $N = 2$ or $N = 3$. However, compared with Proposition 1, an important point of Proposition 2 is that the gain F is not coupled with $P, Q_N, S_1, R_1, R_2, L_1$, and L_2 , but with the free-weighting matrices Y_1 and Y_2 , which enables us to solve the control gain F based on Proposition 2.

Remark 6: Propositions 1 and 2 present a novel method for stability analysis of the system (9). This method skillfully builds some connection with the free-weighting matrix approach, looped functional techniques, and the N -order Bessel–Legendre inequalities so that less conservative stability conditions are derived. On the other hand, since Propositions 1 and 2 depend on the integer N , similar to the proof of [41, Th. 1], one can prove that both Propositions 1 and 2 form a hierarchy of linear matrix inequality conditions. That is, for given h_m and F , one has $h_{M,i} \leq h_{M,i+1}$, where $h_{M,i}$ is the upper bound of h_M obtained from Proposition 1 or 2 with $N = i$ ($i \geq 2$). This means that the larger N , the more robustness of the NCS against DoS attacks.

IV. RESILIENT CONTROLLER DESIGN

In this section, we provide an algorithm to design the control gain F based on Proposition 2.

Proposition 3: For given two constants h_m and h_M with $h_M \geq h_m > 0$ and a positive integer N , and two scalars ε_1 and ε_2 , the system (1) is stabilizable via the resilient controller (7) with $F = KY_0^{-1}$ if there exist compatible real matrices $\bar{P} > 0, \bar{R}_1 > 0, \bar{R}_2 > 0, \bar{Q}_N, \bar{S}_1, \bar{S}_2 = \bar{S}_2^T, \bar{\mathcal{M}}_1, \bar{\mathcal{M}}_2, \bar{L}_i$ ($i = 1, 2, \dots, 8$), K and Y_0 such that, for $\Delta_k \in \{h_m, h_M\}$

$$\begin{aligned}
\tilde{Y}_1 &\triangleq \begin{bmatrix} \text{He}\{Y_0\} + \Delta_k \bar{R}_1 & \hat{\Psi}_{01} + \hat{\Psi}_{02} + \Delta_k \hat{\Psi}_{11} & 0 \\ \star & \hat{\Psi}_{03} + \hat{\Psi}_{04} + \Delta_k \hat{\Psi}_{12} & \Delta_k \bar{\mathcal{M}}_2^T \\ \star & \star & -\Delta_k \bar{\mathcal{R}}_{2N} \end{bmatrix} \\
&< 0 \quad (48) \\
\tilde{Y}_2 &\triangleq \begin{bmatrix} \text{He}\{Y_0\} + \Delta_k \bar{R}_2 & \hat{\Psi}_{01} + \hat{\Psi}_{02} + \Delta_k \hat{\Psi}_{21} & 0 \\ \star & \hat{\Psi}_{03} + \hat{\Psi}_{04} + \Delta_k \hat{\Psi}_{22} & \Delta_k \bar{\mathcal{M}}_1^T \\ \star & \star & -\Delta_k \bar{\mathcal{R}}_{1N} \end{bmatrix} \\
&< 0 \quad (49)
\end{aligned}$$

where $\bar{\mathcal{R}}_{iN} := \text{diag}\{\bar{R}_i, 3\bar{R}_i, \dots, (2N+1)\bar{R}_i\}$ ($i = 1, 2$), and

$$\begin{aligned}
\hat{\Psi}_{01} &:= \bar{P}e_1 + (\bar{S}_1 E_2 - \bar{S}_1^T E_1) C_4 - AY_0 e_1 - BK e_2 \\
&\quad + Y_0^T (\varepsilon_1 e_1 + \varepsilon_2 e_2) \\
\hat{\Psi}_{03} &:= \text{He}\{C_1^T \bar{Q}_N C_4 + \mathcal{C}_1^T \Pi_N^T \bar{\mathcal{M}}_1 + \mathcal{C}_2^T \Pi_N^T \bar{\mathcal{M}}_2 \\
&\quad - (\varepsilon_1 e_1^T + \varepsilon_2 e_2^T)(AY_0 e_1 + BK e_2)\}
\end{aligned}$$

and $\hat{\Psi}_{02}, \hat{\Psi}_{04}$, and $\hat{\Psi}_{ij}$ ($i, j = 1, 2$) are obtained from Ψ_{02}, Ψ_{04} , and Ψ_{ij} ($i, j = 1, 2$) in Proposition 2 by replacing the matrix variables Q_N, S_2 , and L_i ($i = 1, 2, \dots, 8$) with $\bar{Q}_N, \bar{S}_2$, and $\bar{L}_i$ ($i = 1, 2, \dots, 8$), respectively.

Proof: If the matrix inequality (42) is satisfied, $\text{He}\{Y_1\} + \Delta_k \bar{R}_1 < 0$, which means that the matrix Y_1 is invertible.

Introduce a transformation matrix $\mathcal{T}$ as

$$\begin{aligned}\mathcal{T} &:= \{Y_1^{-1}, \mathcal{T}_0, \mathcal{T}_{N+1}\} \\ \mathcal{T}_0 &:= e_1^T Y_1^{-1} e_1 + e_2^T Y_1^{-1} e_2 + e_3^T Y_1^{-1} e_3 + \sum_{j=4}^7 e_j^T \mathcal{T}_N e_j \\ \mathcal{T}_m &:= \text{diag} \left\{ \underbrace{Y_1^{-1}, \dots, Y_1^{-1}}_m \right\}, \quad m > 1.\end{aligned}$$

Then $\mathcal{T}$ is nonsingular. Notice that

$$\begin{aligned}Y_1^{-1} \Psi_{01} \mathcal{T}_0^T &= Y_1^{-1} P Y_1^{-T} e_1 + Y_1^{-1} (S_1 Y_1^{-T} E_2 - S_1^T Y_1^{-T} E_1) C_4 \\ &\quad - (A Y_1^{-T} e_1 + B F Y_1^{-T} e_2) + Y_1^{-1} Y_2^T \mathcal{T}_0^T \\ \mathcal{T}_0 L_7 (\Sigma_N \mathcal{C}_1 - \tilde{\Sigma}_N e_4) \mathcal{T}_0^T &= \mathcal{T}_0 L_7 (\Sigma_N \mathcal{T}_{N+2}^T \mathcal{C}_1 - \tilde{\Sigma}_N \mathcal{T}_N^T e_4) \\ &= \mathcal{T}_0 L_7 \mathcal{T}_{N-1}^T (\Sigma_N \mathcal{C}_1 - \tilde{\Sigma}_N e_4).\end{aligned}$$

Set $Y_2 = \varepsilon_1 Y_1 e_1 + \varepsilon_2 Y_1 e_2$, where ε_1 and ε_2 are two tuning parameters, and let

$$\begin{aligned}\bar{P} &= Y_1^{-1} P Y_1^{-T}, \bar{S}_1 = Y_1^{-1} S_1 Y_1^{-T}, \bar{R}_1 = Y_1^{-1} R_1 Y_1^{-T} \\ \bar{R}_2 &= Y_1^{-1} R_2 Y_1^{-T}, \bar{S}_2 = \mathcal{J}_2 S_2 \mathcal{J}_2^T, \bar{Q}_N = \mathcal{J}_{2N+3} Q_N \mathcal{J}_2^T \\ \bar{L}_1 &= Y_1^{-1} L_1 \mathcal{J}_N^T, \bar{L}_2 = Y_1^{-1} L_2 \mathcal{J}_N^T, \bar{L}_3 = Y_1^{-1} L_3 \mathcal{J}_{N-1}^T \\ \bar{L}_4 &= Y_1^{-1} L_4 \mathcal{J}_{N-1}^T, \bar{L}_5 = \mathcal{T}_0 L_5 \mathcal{J}_N^T, \bar{L}_6 = \mathcal{T}_0 L_6 \mathcal{J}_N^T \\ \bar{L}_7 &= \mathcal{T}_0 L_7 \mathcal{J}_{N-1}^T, \bar{L}_8 = \mathcal{T}_0 L_8 \mathcal{J}_{N-1}^T, \bar{M}_1 = \mathcal{J}_{N+1} M_1 \mathcal{T}_0^T \\ Y_0 &= Y_1^{-T}, K = F Y_1^{-T}, \bar{M}_2 = \mathcal{J}_{N+1} M_2 \mathcal{T}_0^T.\end{aligned}$$

Performing a congruent transformation on Υ_1 and Υ_2 in (42) and (43), it is not difficult to verify that

$$\tilde{\Upsilon}_1 = \mathcal{T} \Upsilon_1 \mathcal{T}^T, \quad \tilde{\Upsilon}_2 = \mathcal{T} \Upsilon_2 \mathcal{T}^T \quad (50)$$

where $\tilde{\Upsilon}_1$ and $\tilde{\Upsilon}_2$ are given in (48) and (49), respectively. Thus, if $\tilde{\Upsilon}_1 < 0$ and $\tilde{\Upsilon}_2 < 0$, then $\Upsilon_1 < 0$ and $\Upsilon_2 < 0$. By Proposition 2, the system (1) is stabilizable under the resilient control (7) with $F = K Y_0^{-1}$. ■

Remark 7: Based on the looped-functional analysis method, Proposition 3 offers a criterion to design stabilizing state feedback controllers for a class of networked systems subject to DoS attacks. If the maximum and minimum duration time of DoS attacks from the *logic* processor belong to a certain range $[h_m, h_M]$, a stabilizing control gain F can be worked out by $F = K Y_0^{-1}$ provided that the linear matrix inequalities (48) and (49) are satisfied. However, in [33] and [34], although some good results on DoS attacks are presented based on DoS parameters (frequency and average dwell time), they can be hardly used to design suitable control gains even for state feedback control.

V. APPLICATION TO SATELLITE CONTROL SYSTEM

Consider the physical plant in Fig. 1 to be a satellite control system shown in Fig. 3, from which the motion equations of

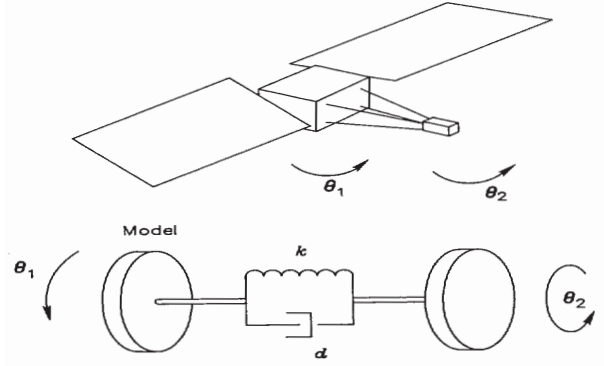


Fig. 3. Sketch of the satellite control system.

the satellite system can be obtained as [44]

$$\begin{cases} J_1 \ddot{\theta}_1 + d(\dot{\theta}_1 - \dot{\theta}_2 + \kappa(\theta_1 - \theta_2)) = T_c \\ J_2 \ddot{\theta}_2 + d(\dot{\theta}_2 - \dot{\theta}_1 + \kappa(\theta_2 - \theta_1)) = 0 \end{cases} \quad (51)$$

where κ and d are torque constant and viscous damping constant, respectively; J_1 and J_2 are inertias and T_c is the control torque. Let $x = \text{col}\{\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2\}$ and $u = T_c$. Then the state equations can be described as (1) with

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{\kappa}{J_1} & \frac{\kappa}{J_1} & -\frac{d}{J_1} & \frac{d}{J_1} \\ \frac{\kappa}{J_2} & -\frac{\kappa}{J_2} & \frac{d}{J_2} & -\frac{d}{J_2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{J_1} \\ 0 \end{bmatrix}.$$

Set $\kappa = 0.09$, $d = 0.0038$, and $J_1 = J_2 = 1$. Then it is easy to verify that A is unstable. We take this practical system to show the effectiveness of the proposed method.

Vulnerability to DoS Attacks: For the satellite control system with no communication networks, using the method similar to [45], a stabilizing controller $u(t) = Fx(t)$ can be designed with

$$F = [-3.3120 \quad 0.4845 \quad -13.4385 \quad -3.48]. \quad (52)$$

Under this controller, it is found that this system is *vulnerable* to DoS attacks. Suppose that the maximum duration time of each DoS attack is greater than 0.15 s, that is, $h_M \geq 0.15$. Then the stability of the closed-loop satellite system may be destroyed no matter what the sampling period T_0 is chosen. To show that, Fig. 4 depicts the initial state responses for the worst case where the system is attacked by DoS adversaries periodically with $h_m = h_M = 0.15$ s. Thus, in this situation, the approaches in both [33] and [34] fail to keep the closed-loop system stable since one cannot design a suitable sampling period or a sampling logic such that the effects of DoS attacks on the system can be reduced.

Resilient Controller Design: From the above analysis, one can see that the closed-loop satellite system with (52) is vulnerable in the presence of DoS attacks if the duration time of each DoS attack is longer than 0.15 s. However, we can take an action to eliminate the effects of DoS attacks on the stability of the system based on information of DoS attacks that the *logic* processor receives (see Fig. 1). In the sequel, we set the sampling period T_0 to be 0.01 s.

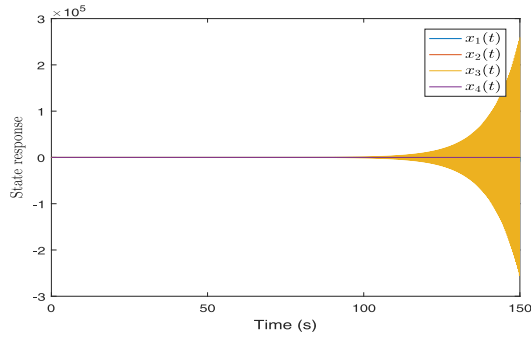


Fig. 4. Initial state responses of the closed-loop satellite system ($h_M = 0.15$ s).

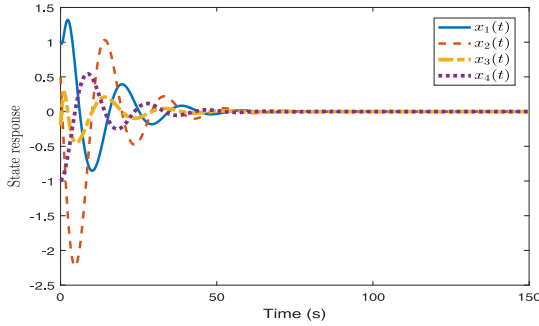


Fig. 5. Initial state responses of the closed-loop satellite system with the control gain (53) ($h_M = 0.5$ s).

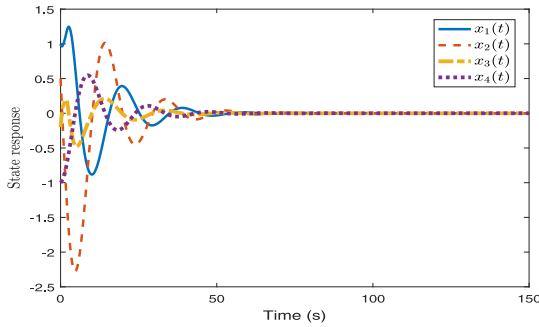


Fig. 6. Initial state responses of the closed-loop satellite system with the control gain (54) ($h_M = 1$ s).

- 1) Suppose that the satellite control system is subject to DoS attacks with the duration time greater than 0.3 s and less than 0.5 s. Then the closed-loop system (9) with (52) is completely destabilized. In this situation, we can exploit Proposition 3 to *actively* design a resilient controller against such DoS attacks. By setting $h_m = 0.3$, $h_M = 0.5$, $\varepsilon_1 = \varepsilon_2 = 1$, and $N = 2$, solving the LMIs (48) and (49), the control gain F can be calculated as

$$F = [-0.9167 \quad -0.0852 \quad -1.4578 \quad -1.6021] \quad (53)$$

Under the control $u(t) = Fx(t)$ with (53), the closed-loop system can preserve the stability even though the maximum duration time of DoS attacks reaches $50T_0$, which can be demonstrated by the initial state responses in Fig. 5 for the worst case.

- 2) Suppose that the satellite control system is subject to malicious DoS attacks with the duration time greater than 0.5 s and less than 1 s. Then by Proposition 3 with $h_m = 0.5$, $h_M = 1$, $\varepsilon_1 = \varepsilon_2 = 1$, and $N = 2$, a stabilizing controller can be designed with the control gain F given by

$$F = [-0.5962 \quad -0.0526 \quad -1.0933 \quad -1.0669]. \quad (54)$$

Under the obtained controller, the initial state responses are depicted in Fig. 6, which shows that this system can retain asymptotic stability even though such malicious DoS attacks try to destroy the system.

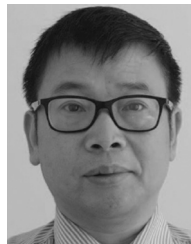
VI. CONCLUSION

The problem of resilient control of an NCS under DoS attacks has been addressed. By introducing a logic processor to capture information about the duration time of each DoS attack, the closed-loop system has been modeled as an aperiodic sampled-data control system. A looped functional method, together with the N -order Bessel–Legendre inequality and the free-weighting matrix approach, has been employed to derive some general criteria to check if the closed-loop stability can be preserved for the NCS under DoS attacks. Moreover, an algorithm has been presented to design resilient state feedback controllers related to the maximum and minimum duration time of DoS attacks. The satellite control system has been given to illustrate the effectiveness of the proposed results.

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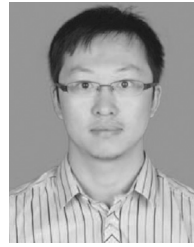
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